IFN647 Week 8 Workshop
Information Needs and Training Data Generation

An **information need** refers to the underlying intent that drives a user to submit a query to an information retrieval (IR) system. However, in practice, queries often provide an incomplete and imprecise representation of this need, as they are typically short and lack sufficient contextual information.

To overcome this limitation, techniques such as query expansion and semi-supervised methods are employed. Instead of expanding individual query terms in isolation, these approaches treat the query holistically to identify semantically related terms. A widely adopted method is **pseudo-relevance feedback (PRF)**. PRF uses the initial query to retrieve a set of top-ranked documents (e.g., the top-k documents or those exceeding a relevance score threshold) based on a given IR model. It then assumes these documents are relevant and extracts informative terms from them to refine and expand the original query. Importantly, this process does not require additional user input.

For example, a user seeking information about crimes committed by individuals who were previously convicted and later released or paroled may issue a short query such as: $Q =$ "convicts, repeat offenders".

Workshop Objectives:

1. Understand how to capture user information needs as the underlying drivers of queries.
2. Understand how to design an algorithm based on the BM25 IF model.
3. Generate training data using the Pseudo-Relevance Hypothesis (i.e., "top-ranked documents are assumed to be relevant") and evaluate the resulting performance.

Task 1. **Ranking documents using the initial query *Q*.**

Design a BM25 based IR model which uses query *Q* to rank documents in the "Training_set" folder and save the result into a text file; e.g., BaselineModel_R102.dat; where each row includes the document number and the corresponding BM25 score or ranking (in descendent order).

Formally describe your idea in an algorithm.

PS: We believe everyone in the class should know how to do text pre-processing and calculate BM25 scores; therefore, we don't need to discuss the details of these parts when you describe your ideas in an algorithm.

Task 2. Training data generation

Pseudo-Relevance Hypothesis: "top-ranked documents (e.g., documents' BM25 scores are great than 1.00) are possible relevant".

Design a python program to find a training set $\underline{D}$ which includes both $\underline{D}^+$ (positive – likely relevant documents) and $\underline{D}^-$ (negative – likely irrelevant documents) in the given un-labelled document set *U*.

The input file is "BaselineModel_R102.dat" or BaselineModel_R102_v2.dat, and
The output file name is "PTraining_benchmark.txt", which has the following sample output:

Please note that this week we do not require you to use BM25 to generate rankings. I used my python program to generate "BaselineModel_R102.dat" using the original BM25 equation (where R=r(t) =0), but it contains some negative BM25 values because N < (2n(t) +1) for some term t. To fix the negatives, I just replaced N with 3N (you could also use the method mentioned in week 7 lecture notes for the negative score issue in BM25), then we can generate "BaselineModel_R102_v2.dat".

```
R102 73038 1
R102 26061 1
R102 65414 1
R102 57914 1
R102 58476 1
R102 76635 1
R102 12769 1
R102 12767 1
R102 25096 1
R102 78836 1
…
R102 86912 0
R102 86929 0
R102 11922 0
R102 14713 0
R102 11979 0
…
```

Task 3. Discuss the precision and recall of the pseudo-relevance assumption.

Compare the relevance judgements (the true benchmark – "Training_benchmark.txt") and the pseudo one ("PTraining_benchmark.txt") and display the number of relevant documents, the number of retrieved documents, the recall, precision and F1 measure by using pseudo-relevance hypothesis, where we view documents in $\underline{D}^+$ as retrieved documents). You may use the function you designed in week 7). The following are possible outputs:

```
the number of relevant docs: 106
the number of retrieved docs: 25
the number of retrieved docs that are relevant: 20
recall = 0.18867924528301888
precision = 0.8
F-Measure = 0.30534351145038174
```

Please analyse the above outputs to understand possible uncertainties (or errors) in the generated training set “PTraining_benchmark.txt”. You can also discuss any possible ways to generate a high-quality training set.